

Using deep learning models and FinBERT to predict the stock price of top banks in the Dhaka Stock Exchange

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Abstract— Bangladesh's economy, like many other contemporary nations, is fundamental to its overall structure. The stock market is one crucial and well-known component of this. Therefore, it is essential to anticipate it earlier to better prepare for market swings. The price of stocks is influenced by a wide range of factors, including geopolitical and economic ones. However, stock data and news sentiment are two that we can leverage ourselves in stock forecasting. Since news strongly influences the stock market, regular and institutional investors often base their decisions on news analysis. Integrating news sentiment analysis into a predictive model holds immense potential when building stock price forecasters. In our research, we have built a web application that reads data from the Dhaka Stock Exchange (DSE) and then relays the output prediction to the user. We use LSTM and GRU for the stock forecasting models on seven banks—four private and three public. For the sentiment analysis task for each of these seven datasets, we employ FinBERT. React, Flask, and MongoDB power our web application, which serves as the medium for all communication. Furthermore, we can confidently state that our research outcomes were precise and valuable regarding the future prices of Bangladeshi stocks. This paper will demonstrate and explain this endeavor's process and outcome trajectory.

Keywords- Stock price forecasting; DSE; GRU; LSTM; Finbert.

I. INTRODUCTION

Around 61 percent of adults in the USA invest in the stock market, compared to Bangladesh, where around 100,000-200,000 people out of a population of 170-180 million actively invest, which is less than or near 1 percent [2, 3]. The few fundamental reasons are financial literacy and investing in other asset classes such as real estate and gold. However, the 1 percent number is most likely to increase due to a growing young population who are more aware of financial investments and the growing economy of Bangladesh, which encourages not only local investors but also foreign investors to invest in the stock markets of Bangladesh. Bangladesh has two stock exchanges, the Dhaka Stock Exchange and the Chittagong Stock Exchange, with Dhaka being the primary one. Due to the growing interest in investing in the stock market, development work is crucial. A large portion of regular investors base their decisions on market-related news.

Therefore, to build a tool for regular investors, it is essential to remember how the news affects the stock market movement. Using historical prices and news sentiment scores, we intend to build a stock predictor predicting the stock price movement of top private banks in the Dhaka stock exchange. We select private banks because they are more established than other industries, and, as a result, their prices are less volatile for sentiment scores of news. We chose the Finbert model [11] for its state-of-the-art performance generating sentiment scores for financial-related news. Then, we used deep learning models such as LSTM and GRU to build a stock predictor on top private banks in the Dhaka stock exchange. As news plays a significant role in influencing investors' emotions, building a stock predictor that takes account of the sentiment score of market news and historical prices is likely to give better results.

II. LITERATURE REVIEW

In the financial market, predicting stock prices is a top goal. News is pivotal in influencing investor sentiments, affecting emotions from fear to optimism. While positive news indicates corporate success, negative news, such as scandals or bad corporate practices, can turn away investors. Understanding the intricate relationship between news, investor sentiments, and stock price is essential for determining corporate practice. Humans are sometimes irrational; most do not make decisions based on mathematical formulas. Thus, building a forecast model using mathematical formulas might give an inaccurate answer. A large number of amateur investors make their financial decisions based on emotion. Despite the challenges of irrationality, leveraging technology can provide investors with valuable insights into the potential impact of news events on market sentiment and stock price. Understanding the relationship between news, sentiment, and stock price can open up new dynamics in the financial market. Sentiment analysis is also known as opinion analysis. Using machine learning for sentiment analysis was first introduced in 2002 [13].

Over there, the sentiments of movie reviews from IMBD were found out. The primary objective of sentiment analysis is to determine whether a text is labeled as positive, neutral, or negative. Sentiment analysis uses a vast range of fields, from understanding the sentiment of customer reviews and the sentiment of the mass population for a political party to predicting oil prices and stock prices. A quasi-linear relationship was shown between oil prices and the sentiment of text on the internet. The stronger the sentiment of the text, the better the forecast of oil prices [14]. On the other hand, T.H. Nguyen [16] pioneered the use of sentiment analysis for large-scale test data to predict stock prices. This indicates that sentiment score plays an important role when forecasting a market. The Stock market operates on a structured framework, generating vast amounts of data daily. This large amount of data makes it a suitable candidate for machine learning algorithms, one such example is to analyze stock markets using LSTM, RNN and CNN-sliding window model [7]. Recently, there has been an interest in using natural language processing (NLP) models to analyze textual information surrounding the financial market. BERT stands out for its effectiveness among these models. Bert is a conceptually simple model, yet it showed powerful results in practice. It obtained new state-of-the-art results on 11 natural language processing tasks [15]. Finbert is a domain-specific Bert model pre-trained through multi-task learning on financial data [12]. The Finbert model gives out sentiment scores of financial-related news, and thus can be used for financial sentiment analysis [10]. Dhaka Stock Exchange started its journey in 1954, previously known as the East Pakistan Stock Exchange Association. It has over 650 securities, including stocks, bonds, and mutual funds [6] and an estimated 72.1 billion USD market capitalization (15th May 2024) [8]. The Dhaka Stock Exchange is smaller and more volatile than the New York Stock Exchange. At the same time, the Dhaka Stock Exchange has a market size of billions of dollars, while the New York Stock Exchange has a market size of trillions of dollars. Due to the minor nature of the Dhaka stock market in Bangladesh, it has witnessed less development compared to established stock markets; however, due to the growing economy of Bangladesh, more investors are willing to invest in Bangladesh. Not much work has been done on the Dhaka stock exchange compared to the New York stock exchange, thus leaving room for innovation. Building a stock predictor for listed companies in the stock market is likely to assist investors in making their purchase decisions. Due to the volatile nature of the stock market, it is more likely to be affected by the news; thus, building a predictor based on historical price and sentiment scores of news is most likely to give better results. In existing works, Hasan et al. [9] used deep learning models such as LSTM to predict the closing price in the Dhaka stock exchange, remarkably the closing price of Grameenphone and ACI. This indicates that work has been done on the Dhaka stock exchange based on historical prices.

However, there needs to be more information on the relationship between the Dhaka stock market and its corresponding market news. Most of the general investors base their decision-making on seeing market news. Human beings are known to make decisions based on emotion; thus, understanding the market sentiment is crucial to understanding how the stock price will move. One of the great ways to understand stock market sentiment is from stock market-related news. Good news from a corporation encourages investors to buy that stock, while lousy news discourages them from buying or selling it. For our research, we intend to use the Finbert model to find out sentiment scores of market-related news and then establish a relationship between stock market movement and news sentiment score along with the historical stock price.

III. METHODOLOGY

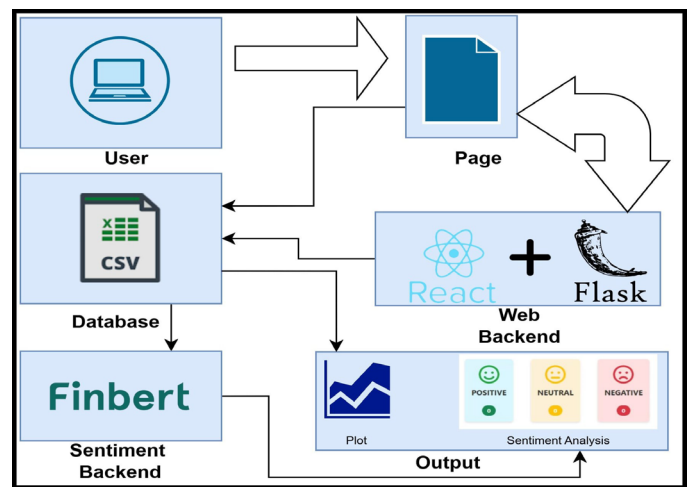


Fig. 1. Workflow diagram for applying LSTM, GRU, and FinBERT to predict stock prices and sentiment scores for stocks from the Dhaka Stock Exchange

A. Methodology

Our objective has two parts. Firstly, we aim to utilize deep learning techniques for our sentiment analysis of specific news based on their news using Finbert. Secondly, we intend to forecast trends on their stock market data using GRU and LSTM. By merging datasets for stock and news data of the same bank, we can simultaneously conduct sentiment analysis and forecasting tasks. We will start by combining the dataset and carrying out standard preprocessing. Next, we will identify which features are necessary for sentiment analysis and which are needed for forecasting. Then, we will train our models with a dataset to perform these operations effectively.

First, we collected data from the news archive section of the Dhaka Stock Exchange [22]. This archive section contains all the news about a particular stock listed on the Dhaka Stock Exchange. Once the news is collected, we then preprocess the news and use the Finbert model to calculate the sentiment scores of the news.

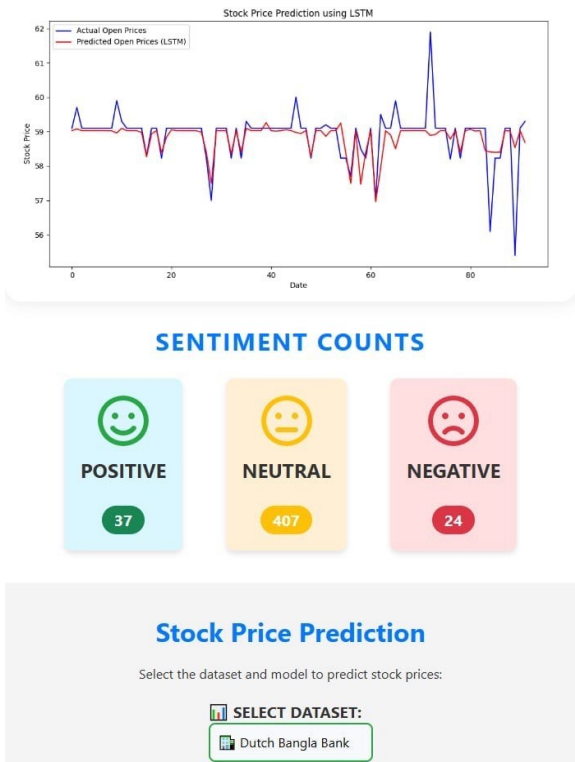


Fig. 2. Output page for web app

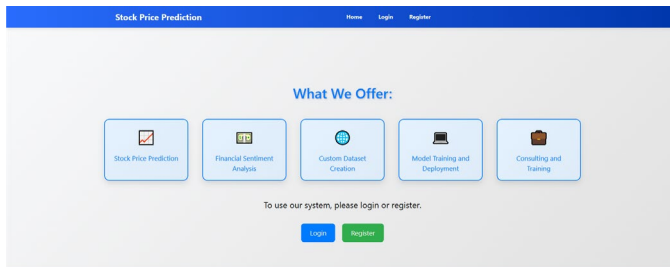


Fig. 3. Landing page for web app

The Finbert model takes a financial text and displays the output in positive, negative, and neutral probability. Those data were stored in a new dataset. In the second part, we collected the historical stock price data from investing.com [21] based on the companies for which we collected news sentiment scores. The sentiment score and stock price were stored together using date. In the later stage, LSTM and GRU were used to build a predictive output that displays the stock price of top banks listed in the Dhaka stock exchange. Fig.1 shows the complete workflow of our system.

B. Website

Lastly, we designed a web page that displays the output of the predictive model when a particular company and a deep learning model are selected. The web app also displays sentiment scores along with the predictive model. The web app aims to build a user interface for investors. The user can select the dataset and bank and then output the sentiment and plot. Fig. 2 shows the representation of the web page. When looking at our datasets, for example, the Dutch Bangla Bank has had more positive and neutral news overall.

This is also reflected in our program's plotting, as the peaks are higher in value and more frequent, and the crests are lower and less frequent. These two data points paint a fuller picture of stock news outcomes. Our research aims to present these to inform the user of the stock market situation. Fig. 3 also shows our landing page when we first enter the site.

C. LSTM

We constructed a predictive model for stock price forecasting for private banks using LSTM. Initially, we gathered historical stock price data from investing.com, focusing on the top four private banks listed on the Dhaka Stock Exchange: Eastern Bank Limited [17], City Bank [18], Dutch Bangla Bank [19], and Islami Bank [20]. The dataset included essential features such as 'Date', 'Open' (opening stock price), 'Price' (closing stock price, which served as our target variable), 'High' (highest stock price during the trading day), 'Low' (lowest stock price during the trading day), and 'Vol.' (trading volume). To normalize the data, we used a min-max scaler. We used the following hyperparameters for training the data: 50 epochs, the batch size used was 32 for training, a 10 percent validation split to monitor model performance, and an Adam optimizer for weight updates.

D. GRU

While using the GRU model, we used similar hyperparameters for training the data: 50 epochs, 32 batch size us, 10 percent validation split, and Adam optimizer for weight updates.

IV. DATA COLLECTION AND PREPARATION

Our data collection procedure is described pictorially in Fig.4 and illustrated in the following section.

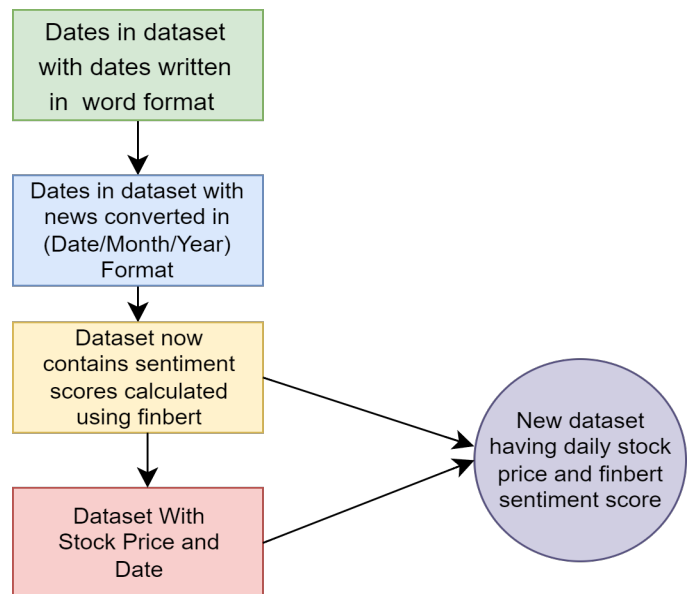


Fig. 4. Representation of data preprocessing

A. Data Collection

For historical stock prices, data was collected from investing.com, a global financial portal that provides new analysis of technical data about the international financial market. Investing.com was selected as data can be downloaded based on a specified time frame, which the original Dhaka stock exchange web portal did not have. The Dhaka Stock Exchange's archive section was used to collect news. The news archive section of the Dhaka Stock Exchange contains stock-related news related to companies listed in the Dhaka Stock Exchange. Amarstock [23] collected news data due to the website's easy interface. Amarstock is powered by the Dhaka stock exchange, thus ensuring the authenticity of the data.

B. Data Preparation

In Table I, the initial dataset was obtained after downloading Dhaka stock market data from investing.com. The features are Date, Price, Open, High, Low, Vol, and change. thousand numbers were separated by a comma, which caused a problem while processing the dataset for our models. The dataset was cleaned to remove the comma after the amount reached a thousand. Table II shows the dataset after removing the commas after thousand.

TABLE I
SHOWS HOW THE STOCK PRICE DATA WAS INITIALLY

Date	Price	Open	High	Low
05/02/2019	1,870.58	1,870.58	1,870.58	1,870.58

TABLE II
SHOWS THE HOW THE STOCK PRICE DATA AFTER PREPROCESSING

Date	Price	Open	High	Low
05/02/2019	1870.58	1870.58	1870.58	1870.58

Similarly, the dataset obtained from Amarstock had to be cleaned. The data was stored in (Name of the weekday, Date, Month in words, and Year) format. The date format in AmarStock had to be converted into a constant format. The emphasis on converting time format was placed on having a similar time frame for the news and historical datasets, as both datasets were merged into a single one based on their date using the datetime library date format. In the next stage, using the Finbert model, the sentiment score of the news was found. The Finbert model provided the probability of positive, neutral, and negative scores for our data collection. Table III shows the new dataset's example and the sentiment score.

TABLE III
NEWS DATASET WITH SENTIMENT SCORE

Date	News	Sentiment	Sentiment Score
5/16/2024	Trading of the shares of the company will resume on 19.05.2024.	Neutral	0.954

V. RESULT ANALYSIS

A. LSTM and GRU

TABLE IV
EVALUATION METRIC OF VARIOUS SCORES OF DUTCH BANGLA BANK IN TESTING

Model	MAE	MSE	RMSE	R ²
LSTM(Stock)	0.29	0.39	0.63	0.29
LSTM(Stock + Sentiment)	0.28	0.38	0.62	0.32
GRU(Stock)	0.23	0.34	0.58	0.39
GRU(Stock + Sentiment)	0.24	0.37	0.61	0.34

TABLE V
EVALUATION METRIC OF VARIOUS SCORES OF DUTCH BANGLA BANK IN TRAINING

Model	MAE	MSE	RMSE	R ²
LSTM(Stock)	0.19	0.12	0.34	0.59
LSTM(Stock + Sentiment)	0.12	0.09	0.30	0.68
GRU(Stock)	0.12	0.07	0.27	0.74
GRU(Stock + Sentiment)	0.14	0.09	0.30	0.68

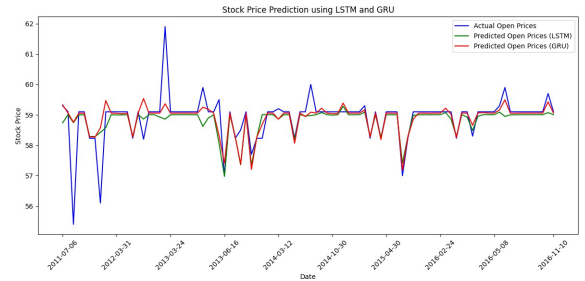


Fig. 5. Result without Finbert

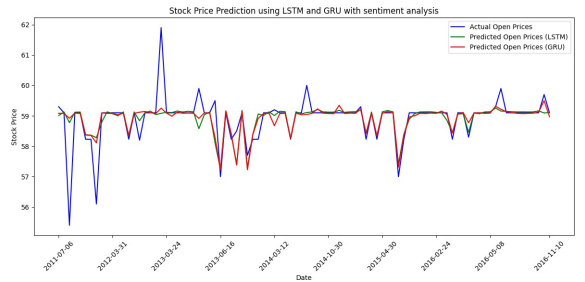


Fig. 6. Result with Finbert

From Tables IV-V, GRU has a marginally lower proportion of errors, as RMSE, MAE, and MSE are much lower. Moreover, GRU also has slightly more predictive power of variations in the dataset, as indicated by the marginally higher R2 value. However, ultimately, they yield similar results in training and testing metrics. The GRU model exhibited better generalization capabilities, as evidenced by the higher R2 score and lower error metrics. This suggests that the GRU model is better suited overall because it exhibits necessary advantageous traits for time series analysis.

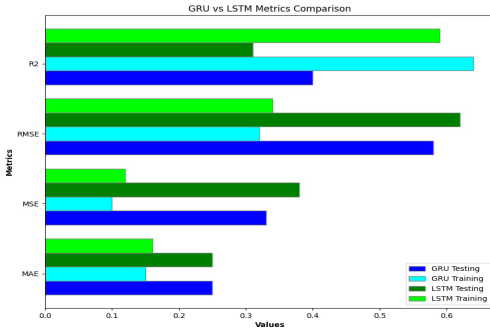


Fig. 7. GRU vs LSTM Metric Comparison

Both models encountered challenges with overfitting, as indicated by the higher performance metrics during training compared to testing. However, the GRU model's simpler architecture mitigates these issues more effectively than the LSTM model. Overall, when we also compare GRU and LSTM evaluation metrics with and without sentiment score, it performs better on average with sentiment score considered, using hyperparameters related to sentiment analysis in addition to regular stock forecasting to integrate the two [1] [4] [5]. We did the same using FinBERT on our dataset and plotted a graph, as shown in Fig. 5 and Fig.6, for forecasting LSTM and GRU of Dutch Bangla Bank, both with and without using FinBERT sentiment scores. We can see that MSE, MAE, RMSE, and R2 metrics, when evaluating values with sentiment scores, tend to perform better than those without avoiding overfitting while holding the general shape of the graph more. This is likely due to more information being available for our models to train on to give a more accurate picture of the market situation. For example, in the period between 2012 and 2013 of the graphs in Fig. 5, we see without sentiment score that the predicted LSTM curves erroneously increased despite the actual values decreasing at that period, with the reverse problem happening for the GRU curve in 2013. Yet, in Fig. 6, this decrease and increase are more accurately predicted. For disparate, unknown spikes and drops, we can thus see that forecasting with the knowledge of sentiment scores can help predict them before they happen. For GRU, we also see that sentiment scores more closely match the actual line than without. The sentiment curve also tends to be less erratic and smoother than the curve without; this is possibly again due to sentiment data acting as a counterbalance to stock data, particularly any sharp decrease or increase in data, which, with sentiment, more context is provided to inform the model that there is a trending increase/decrease within that period before it happens.

When finding out the sentiment score using Finbert, most of the score is shown as neutral. One of the main reasons might be that professionals write the financial text found in the Dhaka stock exchange news archives; they try to avoid writing in a biased tone that might influence the market, thus choosing to state the information by writing in a neutral term. Fig. 8 displays the result.

B. Finbert

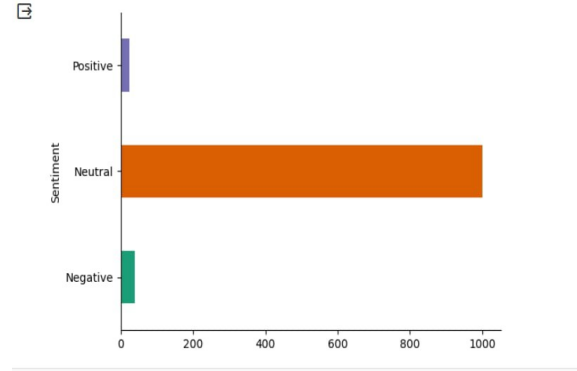


Fig. 8. Sentiment scores from finbert

C. Correlation Matrix

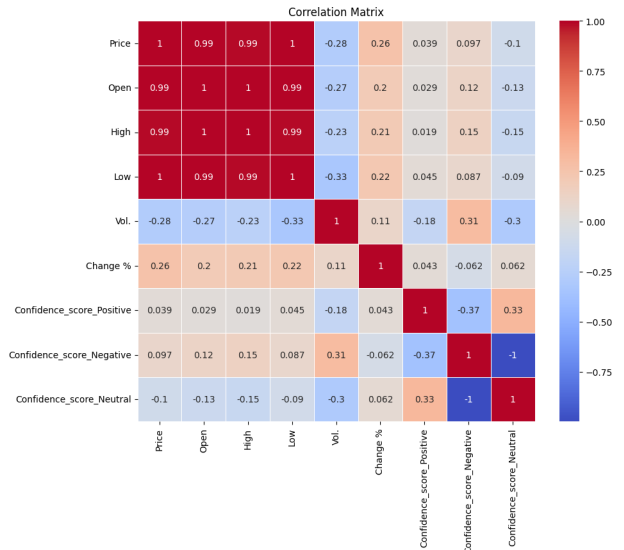


Fig. 9. Correlation Matrix Example

The correlation matrix quantitatively measures the linear relationship between pairs of features in the dataset. Correlation values range from -1 to 1, where 1 indicates a perfect positive correlation, -1 indicates a perfect negative correlation, and 0 indicates no correlation. The correlation matrix of our dataset reveals several notable relationships between the features. The closing price exhibits very high positive correlations with the opening price (0.994), the highest price (0.991), and the lowest price (0.997), indicating that these values are closely aligned throughout the trading day. In contrast, the volume of trades shows a moderate negative correlation with the lowest price of the day (-0.334), suggesting that higher volumes may exert downward pressure on prices.

Furthermore, sentiment analysis scores show exciting correlations. As expected, the positive and negative sentiment scores are moderately negatively correlated (-0.371). Notably, the near-perfect negative correlation between negative and neutral sentiment scores (-0.999) highlights an almost inverse relationship, where decreases in neutral sentiment confidence mirror increases in negative sentiment confidence. These findings suggest that while price movements are closely related to daily trading values, trading volume, and market sentiment, negative sentiment plays a crucial role in influencing market dynamics. These insights provide a deeper understanding of the interplay between trading activities and market sentiment.

VI. CONCLUSIONS

As the market has context behind every curve and line, analyzing stocks and market trends using only these numerical values still needs to be completed. So, this research seeks to further the ongoing task of stock forecasting in the Bangladesh stock market by adding sentiment analysis. We also utilize deep learning methods in our stock market analysis rather than solely traditional economic analysis. Our research thus seeks to further two crucial tasks: developing AI techniques in the Bangladeshi context and analyzing the Bangladesh stock market. Both are underdeveloped compared to the global counterparts of these same fields and will have potential technological and economic benefits of higher investment and more significant innovation if developed further, which induces prosperity. We successfully utilize LSTM, GRU, and FinBERT for our task, utilizing historical data of both stock data and their respective news data for training and evaluation. Pre-processing, training, testing, splitting, and merging were done as necessary. ProsusAI's FinBERT Python library was used for the sentiment analysis task with the news database. Sentiment analysis and time series forecasting were thus our main two tasks to be integrated, utilizing quantitative and qualitative data. Our calculations with R2, MSE, MAE, and RMSE were done to ensure we met the performance benchmarks necessary for accurate data forecasting. Hyperparameter tuning and other appropriate changes were done as required to overcome engineering challenges about overfitting and model complexity, which we faced due to often having to deal with small amounts of data that lack details. Nevertheless, we utilized deep learning with all models successfully. We found GRU to be the best overall in giving the best results regarding accuracy and reducing the risk of overfitting.

More data needs to be available for this research. Especially in regards to public banks, which do not have available data for multiple dates, but we have seen this issue with some private banks as well. This limits any stock analysis as our model will be missing crucial information that can significantly alter the integrity of any predictions. Bangladesh's stock market is underutilized and undervalued by the general public compared to other countries, which hurts the country's overall economy. As a result, any future analysis of Bangladesh's stock market must consider this in addition to its limited current influence and future potential.

And lastly, due to our limited computing power, more advanced models and algorithms could not be used, which future work can explore.

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